

Akhilesh Mahajan

University of Waterloo B.Sc. Candidate (2031) | Honours Physics (Co-Op Program) | 4.0 GPA

Driven to investigate fundamental questions in astrophysics via observational data analysis and computational modeling.

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SKILLS

Programming: Python, MATLAB, C++, Java

Python Packages: NumPy, Pandas, Seaborn, Scikit-learn

Software: Excel & Microsoft Suite, AutoCAD, Arduino & Circuit Simulation

Communication: Scientific Report Writing, Research Poster Design, Public Science Outreach

PROJECTS

House Sales in King County, USA — Predictive Modelling²⁰²⁵

- Analyzed 21,000+ residential property records using Python to clean, wrangle, and engineer features for predictive modelling.
- Built and compared five regression models, including Simple/Multiple Linear Regression, Polynomial Pipeline, and Ridge Regression, achieving R^2 improvement through regularization.
- Constructed a scikit-learn Pipeline automating preprocessing and prediction across 11 housing features.

Automobile Sales Recession Impact Analysis — Data Visualization²⁰²⁶

- Designed 10 visualizations (Matplotlib, Seaborn, Folium) analyzing automobile sales trends across six U.S. recession periods, including an interactive choropleth dashboard.

Double-Slit Interference Pattern Simulator²⁰²⁵

- Developed a numerical wave-optics model reproducing fringe spacing dependence on wavelength, slit separation, and distance.
- Incorporated single-slit diffraction envelope and partial spatial coherence effects.

Refracting Telescope — Optical Design & Fabrication²⁰²³

- Designed with first-principles geometric optics calculations. Built with achromatic doublet lenses and adjustable optical alignment.
- Verified performance against diffraction-limited resolution predictions. Observational validation via resolution of the moon.

RESEARCH EXPERIENCE

Canadian Young Physicists' Tournament — National Competitor, Team Captain²⁰²⁵

- Conducted an original experimental investigation of neodymium magnet dynamics on rotating ferromagnetic rods.
- Designed experimental setup to characterize system across controlled parameter sweeps of rotational velocity and friction regime.
- Developed competing theoretical frameworks, evaluated mechanisms using quantitative experimental data fit comparisons.
- Produced formal research manuscript meeting national competition standards. Delivered technical defence before physics faculty.

Diversity in Research Experiences Across Mathematics Institute (DREAM) — Competitive Research Immersion Programme²⁰²⁴

- Engaged in full research workflow: problem formulation → modelling → technical communication under faculty mentorship.
- Presented formal seminar on Lorentz transformations, special relativity, and general relativity mathematical frameworks.
- Collaborated with graduate student mentors and university researchers on modelling approaches and scientific interpretation.

VOLUNTEER EXPERIENCE

Peer Tutoring Programme for Students with Learning Differences — Spearhead & Lead^{2024–2025}

- Provided individualized instruction in physics and mathematics problem-solving. Supported development of analytical reasoning.

Science Club, Don Mills Collegiate Institute — Founder & President^{2023–2025}

- Established and scaled an organization to 20–30 active members. Organized school-wide STEM events with 100+ participants.
- Designed instructional programming in special relativity, wave optics, thermodynamics, and chaos theory.
- Led volunteer team, managing curriculum planning, demonstrations, and logistics.

AWARDS & HONOURS

- TSI Summer Undergraduate Research Award — Honourable Mention, Top ~10% of Applicants (Trottier Space Institute, McGill University), 2026
- Canadian Young Physicists Tournament — 6th Nationally
- OAPT Physics Contest — School Champion, 2024
- Academic Distinctions — Honour Roll (95% Average)
- Varsity Rugby Distinctions — Captain, MVP, 2022–2025

EDUCATIONAL HIGHLIGHTS

- IBM Data Science Professional Certificate — Coursera, 2026: Hands-on training in statistical analysis, machine learning, and data-intensive computing applied to real-world datasets.
- University of Waterloo — Minor in Computing & Astrophysics. Advanced Functions Award, Physics Award (Top Ranked).